

$$\begin{array}{c}
\frac{}{C\mathcal{R}G \vdash C\mathcal{R}G, G \Vdash A, G \Vdash B} \text{identity} \quad \frac{}{G \Vdash A, C\mathcal{R}G \vdash G \Vdash A, G \Vdash B} \text{identity} \\
\frac{}{(C\mathcal{R}G) \rightarrow (G \Vdash A), C\mathcal{R}G \vdash G \Vdash A, G \Vdash B} \text{identity} \quad \frac{}{G \Vdash A \rightarrow B, (C\mathcal{R}G) \rightarrow (G \Vdash A), C\mathcal{R}G \vdash G \Vdash B} \text{identity} \\
\frac{}{(C\mathcal{R}G) \rightarrow (G \Vdash A), C\mathcal{R}G \vdash C\mathcal{R}G, G \Vdash B} \text{identity} \quad \frac{}{(G \Vdash A) \rightarrow (G \Vdash B), (C\mathcal{R}G) \rightarrow (G \Vdash A), C\mathcal{R}G \vdash G \Vdash B} \text{mImplyL} \\
\frac{}{(C\mathcal{R}G) \rightarrow (G \Vdash A \rightarrow B), (C\mathcal{R}G) \rightarrow (G \Vdash A), C\mathcal{R}G \vdash G \Vdash B} \text{shuffleL} \\
\frac{}{C\mathcal{R}G, (C\mathcal{R}G) \rightarrow (G \Vdash A \rightarrow B), (C\mathcal{R}G) \rightarrow (G \Vdash A) \vdash G \Vdash B} \text{implyR} \\
\frac{}{(C\mathcal{R}G) \rightarrow (G \Vdash A \rightarrow B), (C\mathcal{R}G) \rightarrow (G \Vdash A) \vdash (C\mathcal{R}G) \rightarrow (G \Vdash B)} \text{forallL} \\
\frac{}{\forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)), (C\mathcal{R}G) \rightarrow (G \Vdash A) \vdash (C\mathcal{R}G) \rightarrow (G \Vdash B)} \text{exchangeL} \\
\frac{}{(C\mathcal{R}G) \rightarrow (G \Vdash A), \forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)) \vdash (C\mathcal{R}G) \rightarrow (G \Vdash B)} \text{forallL} \\
\frac{}{\forall F.((C\mathcal{R}F) \rightarrow (F \Vdash A)), \forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)) \vdash (C\mathcal{R}G) \rightarrow (G \Vdash B)} \text{forallR} \\
\frac{}{\forall F.((C\mathcal{R}F) \rightarrow (F \Vdash A)), \forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)) \vdash \forall E.((C\mathcal{R}E) \rightarrow (E \Vdash B))} \text{forallR} \\
\frac{}{C \Vdash \Box A, \forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)) \vdash \forall E.((C\mathcal{R}E) \rightarrow (E \Vdash B))} \text{mBoxL} \\
\frac{}{C \Vdash \Box A, \forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)) \vdash C \Vdash \Box B} \text{mBoxR} \\
\frac{}{\forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)) \vdash (C \Vdash \Box A) \rightarrow (C \Vdash \Box B)} \text{implyR} \\
\frac{}{\forall D.((C\mathcal{R}D) \rightarrow (D \Vdash A \rightarrow B)) \vdash C \Vdash (\Box A) \rightarrow (\Box B)} \text{mImplyR} \\
\frac{}{C \Vdash \Box(A \rightarrow B) \vdash C \Vdash (\Box A) \rightarrow (\Box B)} \text{mBoxL} \\
\frac{}{\vdash (C \Vdash \Box(A \rightarrow B)) \rightarrow (C \Vdash (\Box A) \rightarrow (\Box B))} \text{implyR} \\
\frac{}{\vdash C \Vdash (\Box(A \rightarrow B)) \rightarrow ((\Box A) \rightarrow (\Box B))} \text{mImplyR} \\
\frac{}{\vdash \forall w.(w \Vdash (\Box(A \rightarrow B)) \rightarrow ((\Box A) \rightarrow (\Box B)))} \text{forallR}
\end{array}$$